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Pre- and postpartum metabolizable protein supply: I. Effects on feed intake, lactation performance, and metabolic markers in transition dairy cows.

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ABSTRACT

The objective of this study was to investigate the effect of increasing MP supply in the prepartum, postpartum, or both diets on intake, performance, and metabolic indicators. Multiparous Holstein cows ($n = 96$) were assigned to 1 of 4 treatment groups at 28 d before expected calving following a randomized block design. Prepartum diets were formulated to contain either a control (C; 85 g of MP/kg DM) or high (H; 113 g of MP/kg DM) level of estimated MP. From calving to 21 DIM, diets were formulated to contain either a control (C; 104 g of MP/kg DM) or high (H; 131 g of MP/kg DM) level of estimated MP. To control the potential confounding effect of Met and Lys supply, diets were formulated to supply an equal amount at 1.24 and 3.84 g/Mcal of ME in both prepartum diets and 1.15 and 3.16 g/Mcal of ME in both postpartum diets, respectively. The combination of a pre- and postpartum diet resulted in 4 treatment groups: 1) CC ($n = 23$), 2) CH ($n = 24$), 3) HC ($n = 22$), and 4) HH ($n = 23$). A common lactation diet (113 g of MP/kg DM) was fed from 22 DIM to the end of the observation period at 42 DIM. Milk yield and DMI were collected daily, and plasma metabolic indicators (BHB, fatty acids [NEFA], urea nitrogen [PUN], and glucose) were determined twice weekly from -28 to 28 d relative to calving and once weekly from 29 to 42 DIM. Samples with BHB ≥ 1.2 mmol/L between 3 and 10 DIM were considered hyperketonemia events. Milk composition was determined weekly. Milk yield during 1 to 21 DIM was greater in HH (44.7 ± 1.0 kg/d) compared with CC (39.2 ± 1.0 kg/d) and HC (38.0 ± 1.0 kg/d) and milk yield in CH (42.4 ± 0.9 kg/d) was greater than HC, respectively. From 22 to 42 DIM, milk yield was greater in CH (53.3 ± 1.0 kg/d) and HH (54.1 ± 1.0 kg/d) compared with CC (49.6 ± 1.0 kg/d) and HC (49.3 ± 1.0 kg/d). Dry matter intake (% of BW) and concentrations of milk protein, fat, and total solids

were not affected by treatment. Prepartum concentrations of PUN were greater in H compared with C. From 1 to 21 DIM, PUN concentrations were greater in CH and HH compared with CC and HC. From 1 to 21 DIM, glucose concentrations were lower in HH compared with HC and BHB were greater in CH and HH compared with HC. Concentrations of NEFA, as well as the number of hyperketonemia events did not differ by treatment during this time. From 22 to 42 DIM, concentrations of NEFA were greater in HH compared with HC and concentrations of BHB were greater in CH and HH compared with HC. Overall, feeding CH or HH increased lactation performance without altering intake or hyperketonemia events. Results from this study support formulating a fresh diet to reduce the negative MP balance during early lactation. Key Words: Metabolizable protein, transition cow, amino acids, milk production

INTRODUCTION

The demand for AA increases during the transition period to support mammary development, growth of the fetus and gravid uterus, and the onset of lactation (Bell et al., 1995; Bell, 1995). Consequently, cows enter a protein deficit (Bell, 1995; Mann et al., 2016) that can be exacerbated by diet composition and the reduction in DMI. Recently, NASEM (2021) estimated a requirement of 824 g of MP/d for close-up dry cows, but that estimate does not include requirements for mammary growth and colostrogenesis. Because of the uncertainty of prepartum MP requirements for health and productivity, and the variability observed in DMI, authors have recommended feeding a diet resulting in a prepartum MP supply above 1,000 g/d (Ji and Dann, 2013; Van Saun and Sniffen, 2016; Nydam et al., 2017). However, early lactation milk production responses to different prepartum feeding regimens have been inconsistent (Lean et al., 2013; Underwood et al., 2022; Zang et al., 2022). Farahani et al. (2017) reported increasing prepartum MP from 849 to 1,200 g of MP/d estimated using NRC (2001) resulted

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in increased DMI and milk yield during the first 3 wks of lactation. In Zang et al. (2022), feeding a prepartum MP supply of 1,681 vs. 1,264 g of MP/d estimated using the Cornell Net Carbohydrate and Protein System (CNCPS; Van Amburgh et al., 2015) resulted in lower milk yield and ECM through 12 wk in milk. In a recent meta-analysis, Husnain and Santos (2019) concluded that feeding diets at 1,200 g of NRC (2001) estimated MP/d to multiparous cows during the prepartum period had limited benefits beyond increased milk protein yield (1.15 vs. 1.13 ± 0.03 kg/d) compared with feeding 900 g of MP/d. Moreover, Farahani et al. (2019) reported an interaction between pre- and postpartum MP supply for DMI and milk yield further complicating our understanding of an optimum prepartum MP supply.

In contrast to data collected on the response to increasing prepartum MP supply, several researchers have demonstrated that cows respond favorably to an increased MP supply in the immediate postpartum period based on increased early lactation production (Amanlou et al., 2017; Carder and Weiss, 2017; Tebbe and Weiss, 2021). Cows abomasally infused until 29 DIM with an amount of casein designed to fill the estimated protein deficit showed elevated milk, protein, and lactose yields compared with cows infused with water (Larsen et al., 2014). Additionally, cows fed 2,513 vs. 2,227 g of CNCPS estimated MP/d for 3 weeks postpartum had increased ECM, protein, and lactose yields (Zang et al., 2021). However, because aforementioned studies did not balance treatments for Met and Lys supply, it remains unclear whether these responses were mediated by a difference in the supply of these 2 AA rather than the overall supply of estimated MP (Socha et al., 2005; Batistel et al., 2017; Fehlberg et al., 2020). In addition, it is currently unclear if short-term MP supplementation during the prepartum, postpartum, or during both periods can affect lactation performance beyond the initial feeding period. Therefore, we designed an experiment representing 4 periparturient feeding strategies differing in MP supply. We hypothesized that increasing the pre- and postpartum MP supply would synergistically alter DMI, lactation performance, and the metabolic adaptation to lactation. As such, the objective of this study was to compare the effects of increasing the MP supply in the prepartum, postpartum, or both diets on DMI, production of milk and milk components, and concentrations of blood metabolic markers.

MATERIALS AND METHODS

Animals, Feeding, and Management

All procedures were approved by the Cornell University Institutional Animal Care and Use Committee (protocol number 2019–0031). Holstein cows ($n = 96$)

entering the second or greater lactation were enrolled in a randomized block design between May and November 2021. All animals were moved from a far-off dry cow pen to sawdust bedded tie stalls with individual feed bins between 35 and 42 d before expected calving and fed a far-off diet. At 28 d before expected calving, cows were blocked by expected calving date and balanced for parity and previous lactation 305-d mature equivalent milk production. Animals were randomly assigned within block to 1 of 4 dietary treatment groups consisting of a combination of pre- and postpartum diets. The close-up rations were formulated to contain either a control (C; 85 g of MP/kg DM) or high (H; 113 g of MP/kg DM) level of MP. Both prepartum diets were formulated to supply Met and Lys at 1.24 and 3.84 g/Mcal of ME, respectively. From calving to 21 DIM, cows were fed a postpartum TMR formulated to contain either a control (C; 104 g of MP/kg DM) or high (H; 131 g of MP/kg DM) level of MP. Postpartum diets were formulated to supply Met and Lys at 1.15 and 3.16 g/Mcal of ME in both groups, respectively. The combination of a pre- and postpartum diet resulted in 4 treatment groups: 1) control-control (CC), 2) control-high (CH), 3) high-control (HC), and 4) high-high (HH), respectively. Cows were fed a common lactation diet from 22 to 42 DIM, except 30 cows (CC $n = 7$; CH $n = 8$; HC $n = 7$; HH $n = 8$) were retained on their treatment diets to enter a separate study objective (data not shown) before being also fed the common lactation diet starting at 27.2 ± 2.1 DIM. Treatment diets were formulated using CNCPS v. 6.5.5 (AMTS.Cattle. Professional v. 4.17.0.0; AMTS LLC; Table 1; Van Amburgh et al., 2015). The close-up and fresh C diets were formulated to meet the CNCPS MP requirement during the close-up period and represent MP concentrations in fresh cow diets on commercial dairy farms (Kerwin et al., 2022), respectively. The targeted MP concentrations of the close-up and fresh H diets were established to exceed the estimated CNCPS MP requirement during the prepartum period and to decrease the MP deficit during the fresh period, respectively. Metabolizable protein concentration was increased in pre- and postpartum H diets using heat-treated soybean meal (Amino Plus, Ag Processing Inc.) and a protein supplement (ProvAA1 Lysine, Perdue Agribusiness). Thereafter, commercially available rumen protected Met (Smartamine M, Adisseo Inc.) and Lys (USA Lysine, Kemin Industries Inc.) were used to balance Met and Lys relative to ME in the prepartum and postpartum diets. Methionine and Lys were increased in postpartum diets to achieve the optimum Met and Lys supply of 1.14 and 3.03 g AA/Mcal ME defined for lactating cows in Higgs (2014). Cows were fed once daily starting at 0630 h and the offered feed was adjusted daily to allow a minimum of 5% refusals.

Samples of the TMR, grain mixes, and forages were collected weekly, and dried using a forced-air oven at 40°C for 96 h to determine DM. Dry matter of diet components were used to make weekly adjustments to the rations. For analysis of chemical composition at a commercial laboratory (Cumberland Valley Analytical Services), samples were ground through a 2-mm screen in a Wiley mill (Arthur H. Thomas). Samples of the TMR were composited in 4-wk intervals, whereas forage samples were composited into a single sample for the length of the study. Composite TMR samples were submitted for prediction of chemical composition by near-infrared reflectance spectroscopy and wet chemistry analysis of minerals (method 985.01, AOAC International, 2000), sulfur (LecoCorp, 2008), and chloride [potentiometric titration with silver nitrate (Metrohm 848 Titrino Plus; Metrohm USA Inc.) following extraction via 0.5% nitric acid]. Forage samples were submitted for NDF digestibility by near-infrared reflectance spectroscopy and analyzed by wet chemistry for DM (Partial dry matter (Goering and Van Soest, 1970); followed by 105°C for 3h), crude protein (method 990.03, AOAC International, 2000), ADF (method 973.18, AOAC International, 2000), aNDF (Van Soest et al., 1991), lignin (Goering and Van Soest, 1970), sugar (Dubois et al., 1956), starch (Hall, 2009), fat (method 2003.05, AOAC International, 2006), ash (method 942.05, AOAC International, 2000), minerals (method 985.01, AOAC International, 2000), sulfur (LecoCorp, 2008), and chloride [potentiometric titration with silver nitrate (Metrohm 848 Titrino Plus; Metrohm USA Inc.) following extraction via 0.5% nitric acid].

Ambient temperature/relative humidity and light intensity of the barn were recorded in 30- and 15-min intervals, respectively for the length of the study with 2 environmental data loggers (HOBO Models MX2202/MX2301A, Onset Computer Corp.) mounted approximately 2.5 m above the ground over the staff surface. Temperature-humidity index (THI) was calculated using the equation $THI = T - \left\{ 0.55 \times \left[0.55 \times \left(\frac{RH}{100} \right) \right] \right\} \times (T - 58)$, where T = ambient temperature (°F) and RH = relative humidity percent (Westhoff et al., 2023).

Animal Sampling, Analysis, and Calculations

Individual daily feed intake was recorded, and DMI was calculated using the diet DM percentage collected weekly. Cows were weighed and BCS was determined weekly by one investigator using a 5 point scale with 0.25 point increments (Edmonson et al., 1989). Newborn calf birth weight (kg) and calving ease score (1-to-5-point scale with 1 = normal calving without assistance, 2 = minimal assistance, 3 = moderate assistance, 4 = difficult

calving with extraction by multiple people, 5 = very difficult calving with veterinary assistance) were recorded by farm personnel. Cows were milked 3 times daily at 0600, 1400, and 2200 h and daily milk yield was used to calculate average weekly milk yield. Milk samples were collected once weekly for 3 consecutive milkings. Samples were stored at 4°C with bronopol preservative and submitted to a commercial laboratory (Dairy One Cooperative Inc.) for analysis of fat, true protein, lactose, total solids, and milk urea nitrogen (MUN) by Fourier Transform Infrared Spectroscopy (method 972.160; AOAC International, 2012) and SCC by optical fluorescence (method 972.160; AOAC International, 2012). Milk composition was calculated using the milk yield at each milking session to weight the daily milk composition. Energy corrected milk yield was calculated as ECM (kg) = $[(0.327 \times \text{milk yield, kg}) + (12.95 \times \text{milk fat yield, kg}) + 7.65 \times \text{milk protein yield, kg}]$; Tyrrell and Reid, 1965]. Linear score was calculated as $\text{Log}_2[(\text{SCC}/100,000) + 3]$. Nitrogen use efficiency (NUE) was calculated as $\text{NUE} = (\text{milk N}/\text{intake N})$, where milk N was calculated as milk true protein divided by 6.38. Dietary supply of MP and ME were estimated weekly for each cow (AMTS.Cattle. Professional v. 4.17.0.0) using the calf birth weight, days carried calf, BW, and weekly DMI for prepartum estimates, and DIM, BW, weekly DMI, as well as milk yield, and milk composition for all postpartum estimates.

Blood samples were collected from the coccygeal vein or artery into evacuated collection tubes (Becton Dickinson) containing 158 USP units of sodium heparin between 0600 and 0630 h and immediately placed on ice. Baseline samples were collected once 29 to 35 d relative to expected calving before assignment to treatment. Subsequently, blood samples were obtained twice weekly from -28 to 28 d relative to calving and once weekly from 29 to 42 DIM. Plasma was harvested by centrifugation at 2,300 x g at 4°C for 20 min, frozen, and stored at -80°C until analysis. Plasma concentrations of BHB were measured with samples warmed to 37°C using a Precision Xtra point-of-care device (Abbott) as previously described (Leal Yepes et al., 2018). Calibration of the Precision Xtra point-of-care meter was tested before every use with Medisense Glucose and Ketone Control Solution (Abbott). Samples obtained from 3 to 10 DIM with BHB ≥ 1.2 mmol/L were defined as an event of hyperketonemia (McArt et al., 2013). Plasma concentrations of glucose, nonesterified fatty acids (NEFA), and urea nitrogen (PUN) were determined in duplicate by enzymatic colorimetric analysis (PGO enzyme preparation, Millipore Sigma; HR Series NEFA-HR (2), Wako Life Sciences; urea nitrogen procedure number 640, Millipore Sigma; Chandler et al., 2022), respectively. In-house samples with high and low concentrations were run on every plate to determine coefficients of variation (CV).

Table 1. Ingredient composition of the diets

Ingredient, % of DM	Far-off	Treatment diets ¹				Lactation ²
		Prepartum		Postpartum		
		Control	High	Control	High	
Conventional corn silage	56.7	45	45	—	—	37.1
BMR corn silage ³	—	—	—	37.3	37.3	12.1
Haylage	—	—	—	17.8	17.8	12.5
Wheat straw	—	26.7	26.7	4.7	4.7	—
Grass hay	18.9	—	—	—	—	—
Close-up TMR refusal ⁴	18.9	—	—	—	—	—
Corn grain	—	—	—	17.2	17.2	18.6
Grain mix ⁵	4.6	—	—	—	—	—
Grain mix ⁶	—	—	—	—	—	10.8
Soybean meal solvent	—	3.5	3.3	3.9	0.9	8.6
Canola meal solvent	—	1.8	1.5	3.8	1.2	—
Amino Plus ⁷	—	0.7	11.0	1.0	11.4	—
ProvAAl Lysine ⁸	—	0.9	2.2	0.6	2.2	—
Smartamine M ⁹	—	0.13	0.04	0.17	0.10	—
USA Lysine ¹⁰	—	0.45	—	0.41	0.02	—
Citrus pulp dry	—	—	—	2.70	1.00	—
Soybean hulls ground	—	10.4	1.0	3.70	—	—
Wheat middlings	—	2.9	1.2	1.6	0.90	—
Energy Booster 100 ¹¹	—	—	—	1.5	1.6	—
Sodium bicarbonate	—	—	—	0.76	0.70	—
Salt white	—	0.21	0.19	0.62	0.65	—
MIN-AD ¹²	—	0.89	0.86	—	—	—
Magnesium oxide	—	—	—	0.33	0.34	—
Magnesium sulfate	—	0.89	0.86	—	—	—
Calcium carbonate	—	1.83	2.23	1.33	1.42	—
Calcium sulfate dihydrate	—	0.78	0.10	—	—	—
Dicalcium phosphate	—	0.75	0.48	—	—	—
Vitamin E mix ¹³	—	—	—	0.01	0.01	—
Monensin ¹⁴	—	0.02	0.02	0.01	0.01	—
Choline ¹⁵	—	0.43	0.41	0.30	0.29	—
Animate ¹⁶	—	1.27	2.48	—	—	—
Vitamin/mineral mix ¹⁷	0.9	—	—	—	—	—
Vitamin/mineral mix ¹⁸	—	0.45	0.43	—	—	—
Vitamin/mineral mix ¹⁹	—	—	—	0.26	0.26	—

¹Prepartum diets, fed starting at 28 d before expected calving, were formulated to contain Met and Lys at 1.24 and 3.84 g/Mcal metabolizable energy (ME), respectively and supply a control (C; 85 g MP/kg DM) or high (H; 113 g MP/kg DM) level of estimated MP. Postpartum diets, fed from 1 to 21 DIM, were formulated to contain Met and Lys at 1.15 and 3.16 g/Mcal ME, respectively and supply control (C; 104 g MP/kg DM) or high (H; 131 g MP/kg DM) level of estimated MP.

²From 22 to 42 DIM, cows were fed a lactation diet that contained Met and Lys at 1.01 and 2.88 g/Mcal metabolizable energy, respectively and supplied 113 g of estimated MP/kg DM.

³Brown mid-rib corn silage

⁴13.6% CP, 26.0% ADF, 39.9% aNDF, 3.36% lignin, 2.3% sugar, 21.5% starch, 3.35% fat, 7.93% ash, 1.08% Ca, 0.36% P, 0.36% Mg, 1.36% K, 0.29% S, 0.33% Na, 0.47% Cl.

⁵Contains 39.9% citrus pulp, 14.3% Amino Plus (Ag Processing Inc.), 7.5% soybean hulls, 7.6% ProvAAI lysine (Perdue AgriBusiness), 6.8% sodium bicarbonate, 6.5% Palmit 80 (Global Agri-Trade Corp.), 5.1% calcium carbonate, 3.2% NaCl, 2.4% dicalcium phosphate, 1.8% AjiPro-L (Ajinomoto Health & Nutrition), 1.6% magnesium oxide, 1.6% trace mineral premix, 1.6% urea, 0.05% monensin (Rumensin 90; Elanco Animal Health), 0.05% vitamin E.

⁶Contains 28.7% Amino Plus (Ag Processing Inc.), 13.1% blood meal, 11.8% sodium sesquicarbonate, 9.4% chocolate dairy mix, 9.0% soybean hulls, 7.3% calcium carbonate, 4.2% bypass fat (Cargill Animal Nutrition), 3.5% salt, 3.0% urea, 2.7% Celmanax SCP (Arm and Hammer), 1.8% mono dicalcium phosphate, 1.7% magnesium oxide 54%, 1.7% potassium magnesium sulfate, 0.50% Smartamine M (Adisseo), 0.5% Diamond V XPC (Diamond V North America), 0.40% selenium 0.06%, 0.26% Zinpro 5 (Zinpro Corp.), 0.17% Dairy ADE (Contains 30,073 kIU/kg vitamin A, 5,783 kIU/kg vitamin D, and 92,534 IU/kg vitamin E; Cargill Animal Nutrition), 0.161% trace mineral mix (Contains 2.959% Ca, 17.503% S, 0.657% Cl, 0.231% Mg, 0.141% Na, 0.128% K, 160,210.53 mg/kg Zn, 142,105.26 mg/kg Mn, 23,684.21 mg/kg Cu, 3,526.32 mg/kg Co, 3,157.90 mg/kg I, and 196.84 mg/kg Fe; Cargill Animal Nutrition), 0.057% monensin (Monovet 90; Huvepharma), 0.05% Vitamin E (Contains 510,750 IU/kg), 0.009% copper sulfate.

⁷Heat-treated soybean meal (Ag Processing Inc.)

⁸Protein supplement (Perdue AgriBusiness)

⁹Methionine protected with pH sensitive coating (Adisseo)

¹⁰Rumen bypass lysine (Kemin Industries Inc.)

Continued

Table 1 (Continued). Ingredient composition of the diets¹¹Prilled fatty acids (Milk Specialties Co.): 98% FA (2.5% C12:0, 28% C16:0, 45% C18:0, 8.3% C18:1, 1.5% C18:2, 0.1% C18:3)¹²Magnesium limestone (Papillon Agricultural Company)¹³Contains 18.6% Ca, 0.61% Mg, and 500,444.2 ppm Vitamin E¹⁴Rumensin 90 (Elanco Animal Health)¹⁵Reashure (Balchem Co.)¹⁶Anionic supplement (Phibro Animal Health Corp.)¹⁷Contains 16.5% Ca, 2.0% P, 20.75% NaCl, 6.6% Na, 6.0% Mg, 26.4 ppm Se, 68 kIU/kg vitamin A, 23 kIU/kg vitamin D3, 1600 g/ton monensin.¹⁸Contains 26.38% Ca, 0.12% P, 0.42% Mg, 0.21% K, 3.17% S, 275 ppm Fe, 17,569 ppm Zn, 1,727 ppm Cu, 5,688 ppm Mn, 55 ppm Se, 97.3 ppm Co, 98.18 ppm I, 2,341.71 kIU/kg vitamin A, 566.68 kIU/kg vitamin D, 22,658.58 kIU/kg vitamin E.¹⁹Contains 28.16% Ca, 0.33% Mg, 0.09% K, 2.32% S, 870 ppm Fe, 18,235 ppm Zn, 3,291 ppm Cu, 15,995 ppm Mn, 132 ppm Se, 550 ppm Co, 390 ppm I, 1,984.5 kIU/kg vitamin A, 551.25 kIU/kg vitamin D, 9,481.5 kIU/kg vitamin E.

The intra- and interassay CV were 2.9 and 7.0 percent for glucose, 4.4 and 9.4 percent for NEFA, and 5.1 and 5.9 percent for PUN, respectively. Concentrations of albumin (Alb), cholesterol (Chol), and total bilirubin (Bili) were determined using commercial reagents (C244, C103, C314; Catachem Inc.) and a chemistry analyzer (CataChemWell-T; Catachem Inc.; Abuelo et al., 2020) on available data from a random subset of 15 cows from each treatment at 3 DIM to calculate a liver health index (LHI; Kerwin et al., 2023) using the following equation:

$$\text{LHI} = \left[\frac{\text{Alb} - \mu_{\text{Alb}}}{\sigma_{\text{Alb}}} \right] + \left[\frac{\text{Chol} - \mu_{\text{Chol}}}{\sigma_{\text{Chol}}} \right] - \left[\frac{\text{Bili} - \mu_{\text{Bili}}}{\sigma_{\text{Bili}}} \right],$$

where μ = the sampling population mean and σ = the sampling population standard deviation.

Analytical Approach

Sample size was determined on the number of animals needed to achieve a separate study objective (data not shown). Statistical analysis was performed in SAS v. 9.4 (SAS Institute Inc.) for 3 separate periods (−28 to −1 d relative to calving, 1 to 21 DIM, and 22 to 42 DIM). Mixed effects ANOVA were conducted in PROC MIXED to explore differences in enrollment BW, dry period length, days on close-up diet, calf birth weight, change in BW, liver health index as well as concentrations of albumin, cholesterol, and bilirubin. Differences in calving ease, calf sex, enrollment BCS, and change in BCS were investigated by chi-squared test in PROC FREQ. Repeated measures ANOVA were performed for outcome variables repeated over time using PROC MIXED with the fixed effects of treatment, time, parity (2 vs. ≥ 3), and the interaction of treatment and time. The models included the random effect of enrollment block and the repeated effect of time with the subject of cow. Baseline covariate measurements were included in all models when available. Four covariance structures were tested for data equally spaced in time (variance component, compound symmetry, autoregressive order 1, Toeplitz)

whereas 2 covariance structures were tested for unevenly spaced repeated data (unstructured, and ante dependence 1), and the model with the lowest Akaike information criterion was selected. The model assumptions of normality and homoscedasticity of the residuals were visually assessed. Plasma concentrations of NEFA, BHB, and PUN were transformed using the natural logarithm to meet the model assumptions. Outliers were tested for using the INFLUENCE statement and removed when Cook's distance > 0.5 . If the interaction term resulted in $P \leq 0.05$, an F-test was generated (SLICE) to determine at which time point the interaction occurred. Tukey's post hoc test was used to adjust for multiple comparisons generated in the lsmeans or SLICE option. Data are reported as LSM $\pm$ SEM or back-transformed LSM and 95% CI. Significance was declared when $P \leq 0.05$.

RESULTS

Description of Study Participants

Median (range) temperature, relative humidity, THI, and light intensity during the study period of May to November were 17.2 (1.7–32.6) °C, 81.9 (31.3–98.9) %, 61.8 (40.3–85.6), and 54 (0–1,964) lux, respectively. All data from one cow in HC were removed due to premature parturition (260 DCC) and associated complications after calving. Postpartum data from 3 additional cows (CC $n = 1$; HC $n = 1$; HH $n = 1$) were removed due to injury ($n = 2$) or uterine torsion ($n = 1$), respectively. A total of 95 and 92 cows were included in statistical analysis for the prepartum and postpartum periods, respectively.

The chemically analyzed composition of pre- and postpartum diets is in Table 2. Predicted MP concentration for prepartum diets were 81, 87, and 115 g of MP/kg DM for the far-off, C, and H diets, respectively. Estimated MP concentration for postpartum diets were 101, 127, and 113 g of MP/kg DM for C, H, and the lactation diets, respectively. Descriptive statistics for the study animals are presented in Table 3. Cows were dry for an average of 60.5 ± 4.5 d and there was no difference between treat-

ments ($P = 0.25$). Calf birth weight ($P = 0.34$), calf sex ($P = 0.56$), calving ease score ($P = 0.53$), and enrollment BCS ($P = 0.84$) also did not differ by treatment. Enrollment BW was 751 ± 59 kg and 750 ± 73 kg ($P = 0.91$) for C and H, respectively. Cows were on the prepartum treatment diet for an average of 27.5 ± 4.4 d with no difference between treatments ($P = 0.38$). The frequency of postpartum health events (retained placenta $n = 1$ [1 x HH], metritis $n = 3$ [1 x CC, CH, HH], clinical hypocalcemia $n = 4$ [1 x CC, CH, HC, HH], displacement of the abomasum $n = 0$, and mastitis $n = 2$ [1 x CC, HH]) were recorded but not analyzed further due to the low number of overall events.

Prepartum Period

Intake, BW, BCS, and supply of MP and ME. Results for the prepartum period are in Table 4 and Figure 1. Cows fed H had a greater prepartum BW (778 ± 3 vs. 768 ± 3 kg; $P < 0.01$) and relative BW gain (4.78 ± 0.45 vs. $3.21 \pm 0.45\%$ of BW; $P < 0.01$) compared with C, respectively. Absolute DMI was greater in H ($P = 0.04$; Figure 1); however, relative DMI (% of BW) did not differ for H and C (1.80 ± 0.03 vs. $1.77 \pm 0.03\%$ of BW; $P = 0.29$), respectively. Body condition score did not differ by treatment ($P = 0.76$). Dietary supply of MP ($1,606 \pm 27$ vs. $1,180 \pm 27$; $P < 0.01$) and ME (29.9 ± 0.5 vs. 28.3 ± 0.5 ; $P < 0.01$) were greater in cows fed H compared with cows fed C.

Metabolic Indicators. Prepartum concentrations of glucose, NEFA, BHB and PUN are in Figure 2. A treatment x time interaction was identified for prepartum PUN concentrations ($P = 0.03$) such that PUN concentration was greater and changed over time for H but did not change over time in C. Concentrations of NEFA increased steadily ($P < 0.01$) whereas concentrations of glucose decreased during the prepartum period ($P < 0.01$). A treatment x time interaction was detected for glucose ($P = 0.02$), but NEFA ($P \geq 0.24$) and BHB ($P \geq 0.70$) were not affected by treatment or treatment x time.

Postpartum Period.

Intake, BW, BCS, and supply of MP and ME Dry matter intake increased ($P < 0.01$) over time but did not differ by treatment from 1 to 21 DIM ($P = 0.10$; Table 5, Figure 3). Treatment x time interactions were identified for DMI and relative DMI (% of BW) from 22 to 42 DIM ($P < 0.01$) such that relative DMI increased in all treatments from wk 4 to 5 and wk 5 to 6 ($P < 0.02$), except relative DMI did not increase from wk 5 to 6 in CC and HH ($P > 0.21$). Cows fed CH had a lower relative loss of BW (% of BW) compared with CC during 1 to 21 DIM (-4.06 ± 0.60 vs. $-7.36 \pm 0.65\%$ of BW; $P <$

0.01). Body weight remained stable in all treatments during 22 to 42 DIM ($P = 0.16$). Body condition score was not affected by treatment from 1 to 21 DIM ($P = 0.54$) but was higher in CH compared with CC from 22 to 42 DIM ($P = 0.02$). Change in BCS did not differ between treatments during 1 to 21 ($P = 0.42$) or 22 to 42 DIM ($P = 0.98$). Body condition score increased by 0.25 ($n = 1$; 1.1%), did not change ($n = 29$; 31.5%), decreased by 0.25 ($n = 48$; 52.2%), decreased by 0.50 ($n = 13$; 14.1%), or decreased by 0.75 ($n = 1$; 1.1%) points during 1 to 21 DIM. The estimated dietary supply of MP was greater for cows fed CH or HH compared with CC or HC ($P < 0.01$) during 1 to 21 DIM. Estimated supply of ME was greater in CH (59.7 ± 1.1 Mcal/d) compared with CC (53.6 ± 1.1 Mcal/d; $P = 0.03$) and HC (53.5 ± 1.1 Mcal/d; $P = 0.02$) during 1 to 21 DIM. From 22 to 42 DIM, dietary supply of ME did not differ ($P = 0.06$), but cows fed CH ($3,079 \pm 65$ g/d) consumed an elevated MP supply compared with CC ($2,801 \pm 67$ g/d; $P = 0.01$) and HC ($2,798 \pm 68$ g/d; $P = 0.01$).

Milk Production and Composition Results for milk yield and composition are in Table 6 and Figure 4. From 1 to 21 DIM, a treatment x time interaction was identified for milk yield ($P < 0.01$) and ECM ($P < 0.01$) such that cows fed HH had a greater milk yield and ECM at wk 1 compared with all other groups ($P < 0.04$), but milk and ECM were greater at wk 2 and 3 in both CH and HH compared with CC and HC ($P \leq 0.03$). During 22 to 42 DIM, milk yield and ECM remained elevated in cows fed the H postpartum diet compared with cows fed the C postpartum diet ($P \leq 0.04$). During 1 to 21 DIM, concentrations of milk fat ($P = 0.79$), protein ($P = 0.15$), and total solids ($P = 0.90$) did not differ by treatment, but lactose concentration was higher in CH compared with CC (4.81 ± 0.02 vs. $4.76 \pm 0.02\%$; $P = 0.04$), respectively. Milk urea nitrogen was greater in CH and HH compared with CC and HC ($P < 0.01$) from 1 to 21 DIM. During 22 to 42 DIM, concentrations of milk fat ($P = 0.38$), protein ($P = 0.10$), lactose ($P = 0.42$), total solids ($P = 0.45$), and MUN ($P = 0.15$) did not differ by treatment.

Metabolic Indicators Postpartum plasma concentrations of glucose, NEFA, BHB, and PUN are in Table 7 and Figure 5. Glucose concentrations were affected by treatment during 1 to 21 DIM ($P = 0.03$) such that cows fed HH had lower glucose concentrations compared with HC (53.1 ± 1.1 vs. 57.2 ± 1.2 mg/dL; $P = 0.05$). At 24 DIM, glucose concentrations were higher in HC compared with CC ($P = 0.01$), but glucose concentration did not differ from 27 to 40 DIM ($P > 0.44$). Circulating NEFA concentrations were affected by time ($P < 0.01$) but did not differ by treatment from 1 to 21 ($P = 0.41$). From 22 to 42 DIM, NEFA concentrations were higher in HH compared with HC [495 (413–594) vs. 356 (296–428) $\mu\text{Eq/L}$; $P = 0.03$], respectively. Compared with cows fed HC, BHB

Table 2. Average $\pm$ SD nutrient composition of the diets

Component ² (mean $\pm$ SD)	Treatment diets ¹					
	Far-off	Prepartum		Postpartum		Lactation ³
		Control	High	Control	High	
DM, % ⁴	36.7 $\pm$ 6.0	46.4 $\pm$ 3.6	46.3 $\pm$ 3.3	42.6 $\pm$ 2.5	43.1 $\pm$ 2.4	42.5 $\pm$ 1.4
CP, % of DM	13.3 $\pm$ 1.8	11.5 $\pm$ 0.4	14.6 $\pm$ 1.0	14.5 $\pm$ 0.3	15.9 $\pm$ 0.5	15.4 $\pm$ 0.6
aNDF, % of DM	45.6 $\pm$ 3.5	43.4 $\pm$ 1.0	41.3 $\pm$ 0.4	33.3 $\pm$ 0.5	32.0 $\pm$ 0.5	27.6 $\pm$ 0.8
ADF, % of DM	30.5 $\pm$ 1.4	28.2 $\pm$ 0.7	26.0 $\pm$ 0.3	21.7 $\pm$ 0.4	20.2 $\pm$ 0.5	17.4 $\pm$ 0.7
Lignin, % of DM	4.1 $\pm$ 0.2	3.4 $\pm$ 0.2	3.0 $\pm$ 0.3	2.6 $\pm$ 0.2	2.2 $\pm$ 0.2	1.8 $\pm$ 0.2
Sugar, % of DM	4.6 $\pm$ 1.9	3.8 $\pm$ 0.3	4.6 $\pm$ 0.3	4.8 $\pm$ 0.3	5.2 $\pm$ 0.5	5.0 $\pm$ 0.3
Starch, % of DM	15.0 $\pm$ 3.8	21.8 $\pm$ 0.7	20.9 $\pm$ 0.9	24.7 $\pm$ 0.7	24.7 $\pm$ 0.9	32.1 $\pm$ 1.3
Fat, % of DM	3.4 $\pm$ 0.3	3.4 $\pm$ 0.1	2.9 $\pm$ 0.1	4.9 $\pm$ 0.1	4.8 $\pm$ 0.1	4.2 $\pm$ 0.2
Ash, % of DM	7.8 $\pm$ 0.5	7.7 $\pm$ 0.2	7.7 $\pm$ 0.2	8.2 $\pm$ 0.2	8.0 $\pm$ 0.2	6.8 $\pm$ 0.2
Ca, % of DM	1.04 $\pm$ 0.11	1.52 $\pm$ 0.06	1.59 $\pm$ 0.15	1.14 $\pm$ 0.07	1.02 $\pm$ 0.13	0.93 $\pm$ 0.12
P, % of DM	0.33 $\pm$ 0.02	0.43 $\pm$ 0.01	0.40 $\pm$ 0.02	0.39 $\pm$ 0.03	0.36 $\pm$ 0.01	0.44 $\pm$ 0.02
Mg, % of DM	0.34 $\pm$ 0.03	0.45 $\pm$ 0.01	0.51 $\pm$ 0.03	0.43 $\pm$ 0.02	0.41 $\pm$ 0.01	0.32 $\pm$ 0.02
K, % of DM	1.65 $\pm$ 0.46	1.03 $\pm$ 0.03	1.14 $\pm$ 0.05	1.49 $\pm$ 0.12	1.47 $\pm$ 0.05	1.47 $\pm$ 0.05
S, % of DM	0.23 $\pm$ 0.02	0.48 $\pm$ 0.02	0.42 $\pm$ 0.02	0.23 $\pm$ 0.01	0.22 $\pm$ 0.02	0.22 $\pm$ 0.02
Na, % of DM	0.20 $\pm$ 0.09	0.12 $\pm$ 0.02	0.11 $\pm$ 0.02	0.58 $\pm$ 0.04	0.53 $\pm$ 0.02	0.62 $\pm$ 0.11
Cl, % of DM	0.39 $\pm$ 0.04	0.49 $\pm$ 0.03	0.58 $\pm$ 0.04	0.56 $\pm$ 0.02	0.51 $\pm$ 0.03	0.34 $\pm$ 0.03
DCAD, mEq/100 g	25.6 $\pm$ 11.7	-12.1 $\pm$ 2.7	-8.7 $\pm$ 2.3	33.6 $\pm$ 5.1	32.6 $\pm$ 2.8	40.7 $\pm$ 6.3
Estimated Mcal of ME/kg DM ⁵	2.33	2.09	2.13	2.65	2.66	2.69
Estimated g of FA/kg DM ⁵						
C16:0	6.2	4.8	4.3	10.3	10.1	5.3
C18:0	1.0	0.9	0.7	8.8	9.2	2.8
C18:1 Cis	3.9	4.1	4.2	5.9	5.6	5.1
C18:2	9.4	8.2	8.5	10.6	10.4	11.6
C18:3	2.7	2.2	2.1	2.9	2.8	2.5
Total	24.2	21.5	20.8	42.0	41.5	29.8
Estimated MP, g/kg of DM ⁵	81	87	115	101	127	113
Estimated RDP, % of DM ⁵	7.61	7.46	9.04	9.06	8.92	10.46
Estimated RUP, % of DM ⁵	2.37	3.71	7.08	5.00	8.25	6.39
Estimated metabolizable EAA supply, g/d ⁵	541	600	784	1,016	1,296	1,417
Estimated metabolizable NEAA supply, g/d ⁵	538	575	819	1,028	1,389	1,539
Metabolizable AA supply, g of AA/Mcal ME ⁵						
Met	0.85	1.24	1.23	1.18	1.20	1.01
Lysine	2.68	3.88	3.84	3.27	3.31	2.88
Arg	2.19	2.58	3.42	2.28	2.95	2.60
Thr	1.80	2.05	2.54	1.82	2.22	2.00
Leu	2.67	3.18	4.46	2.94	4.01	3.39
Ile	1.86	2.06	2.45	1.84	2.14	2.01
Val	2.07	2.44	3.15	2.14	2.74	2.36
His	0.92	1.18	1.67	1.02	1.45	1.16
Phe	1.74	2.06	2.78	1.84	2.45	2.12
Trp	0.52	0.60	0.68	0.53	0.58	0.57

¹Prepartum diets, fed starting at 28 d before expected calving, were formulated to contain Met and Lys at 1.24 and 3.84 g/Mcal metabolizable energy (ME), respectively and supply a control (C; 85 g MP/kg DM) or high (H; 113 g MP/kg DM) level of estimated MP. Postpartum diets, fed from 1 to 21 DIM, were formulated to contain Met and Lys at 1.15 and 3.16 g/Mcal ME, respectively and supply control (C; 104 g MP/kg DM) or high (H; 131 g MP/kg DM) level of estimated MP.

²Chemical composition represents the mean $\pm$ SD of six 4-wk composite samples.

³From 22 to 42 DIM, cows were fed a lactation diet that contained Met and Lys at 1.01 and 2.88 g/Mcal metabolizable energy, respectively and supplied 113 g of estimated MP/kg DM.

⁴Presented as mean $\pm$ SD of 22 weekly DM measurements.

⁵Estimated in AMTS.Cattle.Professional v. 4.17.0.0 (AMTS LLC) using average dry matter intake for each diet. ME = metabolizable energy, FA = fatty acid.

concentrations were greater in CH and HH during 1 to 21 ($P < 0.01$) and 22 to 42 DIM ($P \leq 0.03$). However, hyperketonemia events in the first 10 DIM did not differ by treatment ($P = 0.13$). Treatment $\times$ time interactions were identified for PUN during 1 to 21 DIM ($P = 0.02$) and 22 to 42 DIM ($P < 0.01$). At 3 DIM, PUN was higher

in cows fed the H postpartum diet compared with CC ($P < 0.01$) and PUN concentration in HH was higher than HC ($P < 0.01$). Concentrations of PUN from 6 to 20 DIM were greater in CH and HH compared with CC and HC ($P < 0.01$). At 24 DIM, PUN concentrations remained elevated in CH and HH compared with CC and HC (P

Table 3. Descriptive statistics (mean $\pm$ SD) for the study animals

Variable	Treatment ¹		<i>P</i> -value ²
	Control (n = 48)	High (n = 47)	
Dry period length, d	60 $\pm$ 4	61 $\pm$ 5	0.25
Days on close-up treatment diet	27 $\pm$ 4	28 $\pm$ 5	0.38
Calf birth weight, kg	44.6 $\pm$ 3.9	43.7 $\pm$ 4.2	0.34
Enrollment BW, kg	751 $\pm$ 59	750 $\pm$ 73	0.91
Enrollment BCS, n (%)			0.84
2.75	1 (2.1)	0 (0)	
3.00	2 (4.2)	2 (4.3)	
3.25	25 (52.1)	23 (48.9)	
3.50	19 (39.6)	20 (42.6)	
3.75	1 (2.1)	2 (4.3)	
Calf			0.56
Female	21 (43.8)	19 (40.4)	
Male	26 (54.2)	28 (59.6)	
Twin	1 (2.1)	0 (0)	
Calving ease score ³			0.53
1	39 (81.3)	41 (87.2)	
2	5 (10.4)	3 (6.4)	
3	4 (8.3)	2 (4.3)	
4	0 (0)	1 (2.1)	
5	0 (0)	0 (0)	

¹Prepartum diets, fed starting at 28 d before expected calving, were formulated to contain Met and Lys at 1.24 and 3.84 g/Mcal metabolizable energy, respectively and supply a control (C; 85 g MP/kg DM) or high (H; 113 g MP/kg DM) level of estimated MP.

²Fixed effect of treatment.

³Calving ease score: 1 = normal calving without assistance, 2 = minimal assistance, 3 = moderate assistance, 4 = difficult calving with extraction by multiple people, 5 = Very difficult calving with veterinary assistance.

< 0.01); however, PUN concentration did not differ by treatment from 27 to 42 DIM ($P > 0.12$). Liver health index did not differ by treatment (CC: 0.24 ± 0.44 ; CH: 0.37 ± 0.44 ; HC: 0.18 ± 0.44 ; HH: -0.79 ± 0.44 ; $P = 0.24$).

Table 4. Effect of increasing prepartum MP supply on prepartum DMI, BW, BCS, and estimated metabolizable energy (ME) and MP supply

Variable ³	Treatment ¹		<i>P</i> -value ²			
	Control (n = 48)	High (n = 47)	Trt	Wk	Parity	Trt x Wk
DMI, kg/d	13.5 $\pm$ 0.3	14.0 $\pm$ 0.3	0.04	<0.01	0.57	0.43
DMI, % BW	1.77 $\pm$ 0.03	1.80 $\pm$ 0.03	0.29	<0.01	0.44	0.90
BW, kg	768 $\pm$ 3	778 $\pm$ 3	<0.01	<0.01	0.92	0.96
BW change, kg	23.9 $\pm$ 3.3	35.1 $\pm$ 3.3	<0.01		0.26	
BW change, %	3.21 $\pm$ 0.45	4.78 $\pm$ 0.45	<0.01		0.17	
BCS	3.39 $\pm$ 0.02	3.39 $\pm$ 0.02	0.76	0.40	0.09	0.35
BCS change, n (%)			0.14			
-0.25	4 (8.3)	2 (4.3)				
0.00	23 (47.9)	33 (70.2)				
0.25	20 (41.7)	12 (25.5)				
0.50	1 (2.1)	0 (0)				
MP supply, g/d ⁴	1,180 $\pm$ 27	1,606 $\pm$ 27	<0.01	<0.01	0.15	0.06
ME supply, Mcal/d ⁴	28.3 $\pm$ 0.5	29.9 $\pm$ 0.5	<0.01	<0.01	0.20	0.38

¹Prepartum diets, fed starting at 28 d before expected calving, were formulated to contain Met and Lys at 1.24 and 3.84 g/Mcal metabolizable energy, respectively and supply a control (C; 85 g MP/kg DM) or high (H; 113 g MP/kg DM) level of estimated MP.

²Fixed effects of treatment (trt), time (wk), parity (2 vs. ≥ 3), and the interaction of treatment x time.

³Reported as LSM $\pm$ SEM.

⁴Dietary supply of MP and metabolizable energy (ME) estimated in AMTS.Cattle.Professional v. 4.17.0.0 (AMTS LLC) weekly using cow's DMI, BW, calf birth weight, and days carried calf.

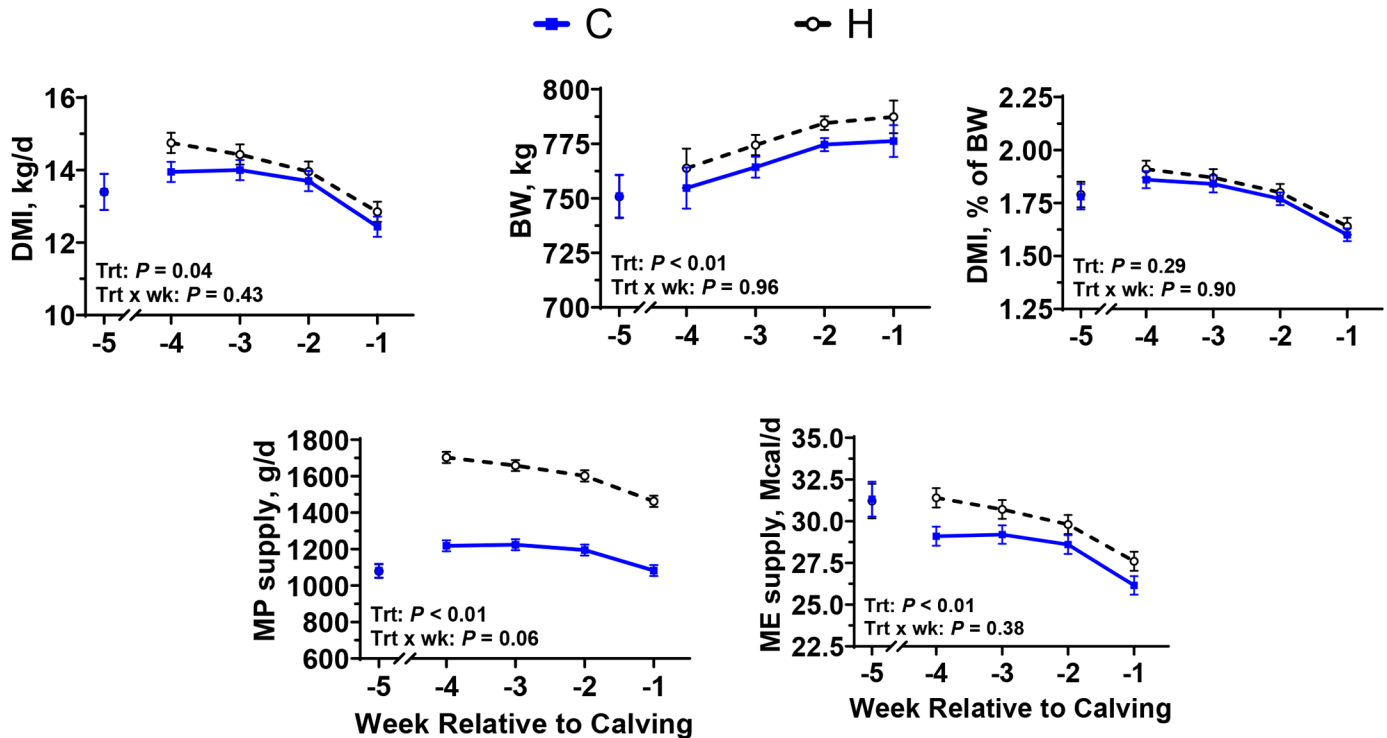


Figure 1. Effect of increasing prepartum MP supply on prepartum DMI, BW, and DMI (% of BW), MP supply, and metabolizable energy supply (ME). Prepartum diets were formulated to contain Met and Lys at 1.24 and 3.84 g/Mcal ME, respectively and supply a control (C; 85 g MP/kg DM) or high (H; 113 g MP/kg DM) level of estimated MP. Metabolizable protein and ME were estimated in AMTS.Cattle.Professional v. 4.17.0.0 (AMTS LLC) weekly using cow's DMI, BW, calf birth weight, and days carried calf. Data presented as LSM $\pm$ SEM *P*-values for all main effects are in Table 4.

DISCUSSION

A successful transition from gestation to lactation has a substantial impact on production, health, and farm profitability (Drackley, 1999) and appears to be influenced, in part, by nutrient availability (Larsen et al., 2014; McCarthy et al., 2015; Cardoso et al., 2020). Advancements in our understanding of the optimum AA supply (LaPierre et al., 2019; Nichols et al., 2019) in tandem with improved diet formulation models (Van Amburgh et al., 2015; NASEM, 2021) have modernized the metric of crude protein toward the use of MP. As such, an opportunity exists to reevaluate protein feeding strategies with a particular focus on the dynamic transition period. Thus, the objective of this study was to compare the effects of increasing the MP supply in the prepartum, postpartum, or both diets on intake, performance, and concentrations of metabolic indicators.

Effect of Two Prepartum MP Levels on Prepartum Outcomes and Calf Birth Weight

In the current study, absolute DMI was greater in cows fed the H prepartum diet, but because of a greater BW

gain in H, relative DMI (% of BW) was not affected by MP supply during the prepartum period. The observed change in BW aligns with hyperplasia of the mammary gland (VandeHaar and Donkin, 1999) and gravid uterus (Bell et al., 1995) during late gestation. Because prepartum MP supply did not affect calf birth weight in agreement with findings from previous authors (Doepel et al., 2002; Farahani et al., 2017; Farahani et al., 2019), the difference in BW gain was not explicit but could be in part explained by an increase in absolute DMI or by differences in labile protein reserves as lean muscle mass that has previously been shown to be affected by the supply of MP fed during late gestation (Swick and Benevenega, 1977; Lean et al., 2013; Van Saun and Sniffen, 2016). Most authors did not detect an effect of MP supply on absolute prepartum DMI (Akhtar et al., 2021; Underwood et al., 2022; Zang et al., 2022); however, in a study by Farahani et al. (2017), prepartum DMI increased when feeding 1,200 or 1,387 g/d compared with 849 g/d of MP estimated using NRC (2001). Rumen-degradable protein is important for optimum rumen digestibility and it was recommended to be supplied at a minimum of 10% of diet DM (NASEM, 2021). Discrepancies in the aforementioned studies might be a result of differences in

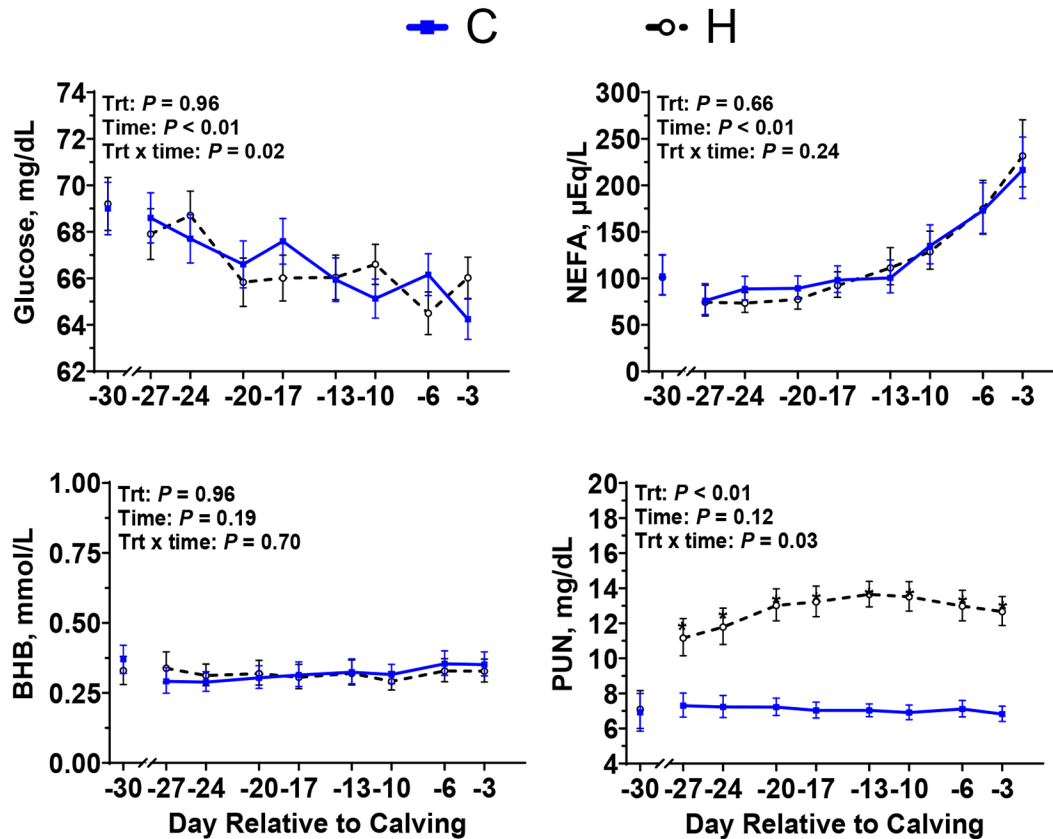


Figure 2. Effect of increasing prepartum MP supply on prepartum concentrations of glucose, nonesterified fatty acids (NEFA), BHB, and plasma urea nitrogen (PUN). Prepartum diets were formulated to contain Met and Lys at 1.24 and 3.84 g/Mcal metabolizable energy (ME), respectively and supply a control (C; 85 g MP/kg DM) or high (H; 113 g MP/kg DM) level of estimated MP. If the interaction term resulted in $P \leq 0.05$, differences ($P \leq 0.05$; Tukey's test) between treatments at individual time point were denoted with *. Data presented as LSM $\pm$ SEM or back-transformed LSM and 95% confidence interval. All models were controlled for the effect of parity $P \geq 0.15$.

RDP, dietary NDF, or a combination of both that affect nutrient digestibility and gut fill. Here, higher concentrations of RDP in the prepartum H diet and elevated BW likely contributed to higher absolute DMI.

Effect of Increasing MP Supply Prepartum, Postpartum, or Both on Postpartum Outcomes

The effect of postpartum MP supply on absolute DMI has yielded mixed results with some authors observing no effect (Carder and Weiss, 2017; Leal Yepes et al., 2019; Tebbe and Weiss, 2021) while other authors reported an increased intake in response to increased postpartum MP or AA supply (Giallongo et al., 2016; Amanlou et al., 2017). Reductions in total-tract NDF digestibility have been reported in Lee et al. (2011) and Lee et al. (2012) when feeding diets deficient in MP suggestive of impaired rumen function due to RDP or ammonia deficiency as discussed above. Furthermore, the magnitude of MP deficiency and the source, palatability, digestibility, AA composition, and AA bioavailability of

feed ingredients used to manipulate MP supply in the aforementioned studies might have affected DMI. With the feedstuffs used and similar estimated RDP in the postpartum diets, results from this study do not suggest postpartum MP supply affects DMI.

Several authors have described positive milk production responses to supplemental AA or MP fed during the close-up (Farahani et al., 2017; Fehlberg et al., 2020) or fresh period (Larsen et al., 2014; Tebbe and Weiss, 2021; Zang et al., 2021), but studies considering the effect of pre- and postpartum MP supply remain scarce (Farahani et al., 2019). Results from this study show that feeding HH increased lactation performance at wk 1 compared with all other treatments and feeding the H postpartum diet increased lactation performance from wk 2 to the end of the observation period. The persistent difference in milk yield in cows fed the H postpartum diet observed beyond the end of the treatment period at 21 DIM contrast with results from authors who only altered postpartum MP supply (Carder and Weiss, 2017; Tebbe and Weiss, 2021; Zang et al., 2021) but might indicate

Table 5. Effect of increasing MP supply in the prepartum, postpartum, or both diets on postpartum DMI, BW, BCS, and estimated metabolizable energy (ME) and MP supply

Variable ³	Treatment ¹				P-value ²			
	CC (n = 23)	CH (n = 24)	HC (n = 22)	HH (n = 23)	Trt	Wk	Parity	Trt x Wk
1 to 21 DIM								
DMI, kg/d	20.2 ± 0.4	21.5 ± 0.4	20.5 ± 0.4	20.7 ± 0.4	0.10	<0.01	0.06	0.68
DMI, % BW	2.98 ± 0.06	3.04 ± 0.06	3.00 ± 0.06	2.94 ± 0.06	0.71	<0.01	0.60	0.52
BW, kg	678 ± 11 ^{bc}	716 ± 11 ^{ab}	676 ± 11 ^c	720 ± 11 ^a	<0.01	<0.01	<0.01	0.05
BW change, kg	-55 ± 5 ^b	-30 ± 5 ^a	-42 ± 5 ^{ab}	-46 ± 5 ^{ab}	<0.01		<0.01	
BW change, %	-7.36 ± 0.65 ^b	-4.06 ± 0.64 ^a	-5.63 ± 0.67 ^{ab}	-6.12 ± 0.67 ^{ab}	<0.01		<0.01	
BCS	3.27 ± 0.04	3.33 ± 0.03	3.31 ± 0.04	3.33 ± 0.04	0.54	<0.01	0.33	0.18
BCS change, n (%)					0.42			
-0.75	0 (0)	0 (0)	0 (0)	1 (4.4)				
-0.50	6 (26.1)	2 (8.4)	2 (9.1)	3 (13.0)				
-0.25	11 (47.8)	11 (45.8)	13 (59.1)	13 (56.5)				
0.00	6 (26.1)	11 (45.8)	7 (31.8)	5 (21.8)				
0.25	0 (0)	0 (0)	0 (0)	1 (4.3)				
MP supply, g/d ⁴	2033 ± 51 ^b	2766 ± 50 ^a	2030 ± 52 ^b	2667 ± 51 ^a	<0.01	<0.01	<0.01	0.20
ME supply, Mcal/d ⁴	53.6 ± 1.1 ^b	57.9 ± 1.1 ^a	53.5 ± 1.1 ^b	56.2 ± 1.1 ^{ab}	<0.01	<0.01	<0.01	0.68
22 to 42 DIM								
DMI, kg/d	25.1 ± 0.5	26.8 ± 0.5	25.1 ± 0.5	25.6 ± 0.5	0.06	<0.01	<0.01	<0.01
DMI % BW	3.86 ± 0.08	3.89 ± 0.07	3.82 ± 0.08	3.77 ± 0.08	0.70	<0.01	0.53	<0.01
BW, kg	650 ± 10 ^b	689 ± 10 ^a	657 ± 10 ^{ab}	681 ± 10 ^{ab}	0.02	0.16	<0.01	0.92
BW change, kg	-0.5 ± 4.3	4.1 ± 4.2	-0.1 ± 4.4	2.2 ± 4.3	0.82		0.62	
BW change, %	-0.1 ± 0.7	0.7 ± 0.6	0.1 ± 0.7	0.4 ± 0.7	0.83		0.66	
BCS	3.04 ± 0.03 ^b	3.17 ± 0.03 ^a	3.08 ± 0.04 ^{ab}	3.10 ± 0.03 ^{ab}	0.03	<0.01	0.88	0.74
BCS change, n (%)					0.98			
-0.50	1 (4.4)	2 (8.3)	1 (4.5)	2 (8.7)				
-0.25	6 (26.1)	7 (29.2)	8 (36.4)	5 (21.7)				
0.00	13 (56.5)	13 (54.2)	10 (45.5)	14 (60.9)				
0.25	3 (13.0)	2 (8.3)	3 (13.6)	2 (8.7)				
MP supply, g/d ⁴	2801 ± 67 ^b	3079 ± 65 ^a	2798 ± 68 ^b	2926 ± 67 ^{ab}	<0.01	<0.01	<0.01	0.01
ME supply, Mcal/d ⁴	67.5 ± 1.4	72.1 ± 1.3	67.5 ± 1.4	69.1 ± 1.4	0.06	<0.01	<0.01	<0.01

^{a-c} LSM with different superscripts differ ($P \leq 0.05$; Tukey's test).¹Prepartum diets, fed starting at 28 d before expected calving, were formulated to contain Met and Lys at 1.24 and 3.84 g/Mcal metabolizable energy (ME), respectively and supply a control (C; 85 g MP/kg DM) or high (H; 113 g MP/kg DM) level of estimated MP. Postpartum diets, fed from 1 to 21 DIM, were formulated to contain Met and Lys at 1.15 and 3.16 g/Mcal ME, respectively and supply control (C; 104 g MP/kg DM) or high (H; 131 g MP/kg DM) level of estimated MP. The combination of a pre- and postpartum diet resulted in treatments: 1) CC, 2) CH, 3) HC, 4) HH. From 22 to 42 DIM, cows were fed a lactation diet that contained Met and Lys at 1.01 and 2.88 g/Mcal ME, respectively and supplied 113 g MP/kg.²Fixed effects of treatment (trt), time (wk), parity (2 vs. ≥ 3), and the interaction of treatment x week.³Reported as LSM $\pm$ SEM.⁴Dietary supply of metabolizable protein (MP) and metabolizable energy (ME) estimated in AMTS.Cattle.Professional v. 4.17.0.0 (AMTS LLC) weekly using cow's DMI, BW, DIM, milk yield, and milk composition.

altered numbers of alveolar cells (Larsen et al., 2014). Mammary cell proliferation and apoptosis was greatest during late gestation and early lactation, respectively (Sorensen et al., 2006; Nørgaard et al., 2008) and can be influenced, in part, by nutrition (Dessauge et al., 2011; Cant et al., 2018; Shabrandi et al., 2019). The current study supports the hypothesis that increasing postpartum MP supply can affect persistency in milk production by means of proliferating and maintaining mammary epithelial cell capacity (Larsen et al., 2014).

Cows fed the H postpartum diet produced a greater total solid yield that was ascribed to increased production of lactose, protein, and fat. Increased production of lactose in response to an elevated MP supply was attributed by others to increase milk yields (Galindo et al., 2015; Cardoso et al., 2021), but the greater glucose demand to

support lactose synthesis in cows fed H postpartum was not accompanied by a greater DMI as discussed above. Theoretically, excess gluconeogenic AA, specifically alanine, could contribute to increased glucose production during this time (Overton et al., 1999; Overton and Waldron, 2004; Larsen and Kristensen, 2013); however, greater glucose appearance in early lactation cows infused with AA have been reported to originate from increased portal rather than hepatic release (Galindo et al., 2015). Simultaneously, elevated concentrations of ketone bodies in the current study, some of which can be synthesized from ketogenic AA, might spare glucose by acting as an alternative energy source in multiple tissues including the mammary gland (Mann and McArt, 2023). Given that postpartum diets had similar concentrations of RDP, increased concentrations of PUN and MUN in

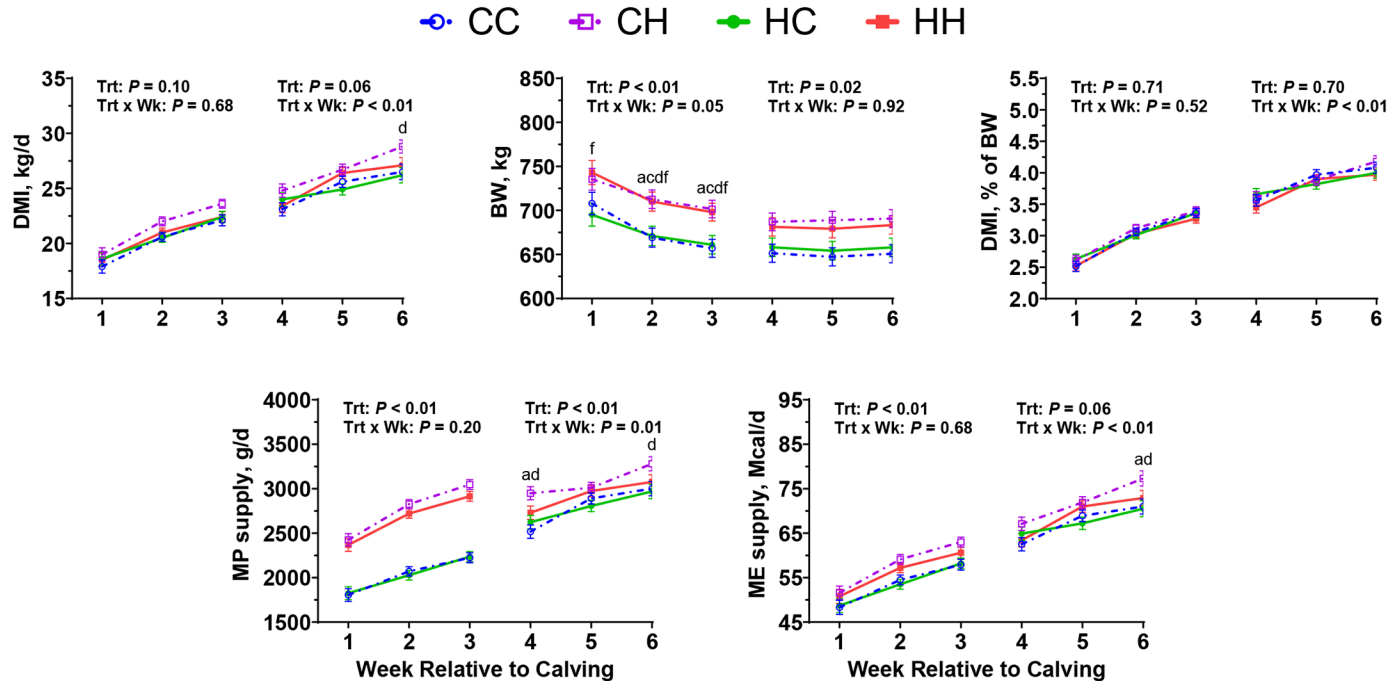


Figure 3. Effect of increasing MP supply in the prepartum, postpartum, or both diets on DMI, BW, DMI (% of BW), MP supply, and metabolizable energy (ME) supply. Prepartum diets were formulated to contain Met and Lys at 1.24 and 3.84 g/Mcal ME, respectively and supply a control (C; 85 g MP/kg DM) or high (H; 113 g MP/kg DM) level of estimated MP. Postpartum diets were formulated to contain Met and Lys at 1.15 and 3.16 g/Mcal ME, respectively and supply control (C; 104 g MP/kg DM) or high (H; 131 g MP/kg DM) level of estimated MP. The combination of a pre- and postpartum diet resulted in treatments: 1) CC, 2) CH, 3) HC, 4) HH. From 22 to 42 DIM, cows were fed a lactation diet that contained Met and Lys at 1.01 and 2.88 g/Mcal ME, respectively and supplied 113 g MP/kg. Metabolizable protein and ME were estimated in AMTS.Cattle.Professional v. 4.17.0.0 (AMTS LLC) weekly using the cow's DMI, BW, DIM, milk yield, and milk composition. If the interaction term resulted in $P \leq 0.05$, letters denote differences ($P \leq 0.05$; Tukey's test) between treatments at individual time point. a = CC vs. CH, b = CC vs. HC, c = CC vs. HH, d = CH vs. HC, e = CH vs. HH, f = HC vs. HH. Data presented as LSM $\pm$ SEM P -values for all main effects are in Table 5.

cows fed the H postpartum diet supports this theory of AA deamination for gluconeogenesis or ketogenesis even though their origin could also be absorption of excess ammonia released in the rumen. Although we were unable to determine to what extent ammonia from the rumen contributed to hepatic ureagenesis, elevated concentrations of PUN were reported in early lactation cows intravenously infused with a mixture of AA (Chandler et al., 2022) or abomasally infused with casein (Larsen et al., 2014), and when cows were fed diets balanced for RDP but exceeding the requirement of RUP (Elrod et al., 1993). Further, previous authors have reported increased MUN concentrations and milk yield when increasing the supply of MP fed postpartum (Tebbe and Weiss, 2021; Zang et al., 2021) and suggest that AA catabolism likely contributed to the magnitude of the production response observed herein.

During the first 21 d, protein yield was greater in cows fed HH compared with CC and HC and protein yield was greater in CH compared with HC in alignment with findings of increased protein yield in response to an elevated postpartum MP supply described previously (Zang et al., 2021). Protein synthesis is responsive to energy as well

as mammary uptake of AA (Daniel et al., 2016; Cant et al., 2018; NASEM, 2021), the latter of which has been shown to increase in early lactation cows abomasally infused with casein (Larsen et al., 2014). The increased protein yield in cows fed the H diet postpartum was supported by a greater MP intake and availability of circulating EAA as shown in our companion paper (Westhoff et al., 2024). Consistent with this observation, feeding diets with lower CP (Law et al., 2009) or MP (Tebbe and Weiss, 2021; Zang et al., 2021) during early lactation has been an effective strategy to increase N efficiency, but often comes at the expense of milk protein yield or of increased muscle mobilization. Although a marker of muscle tissue mobilization, 3-methyl-histidine, did not differ by treatment during the postpartum period as shown in our companion paper (Westhoff et al., 2024), the increase in N use efficiency in CC and HC compared with CH coincided with reduced milk protein yield.

Greater milk fat yield in cows fed the H postpartum diet was driven primarily by greater milk yields. In a study by Chandler et al. (2022), intravenous infusion of a mixture of AA to postpartum cows increased yields of de novo and preformed fatty acids, although preformed

fatty acids were likely due to the difference in DMI leading to greater adipose tissue mobilization. Despite small differences in circulating fatty acids, DMI as well as the loss of BCS and reductions in backfat thickness as described in our companion paper (Westhoff et al., 2024), did not differ by treatment and do not support increased adipose tissue mobilization as the main contributor to increased milk fat yield. Alternatively, direct catabolism of ketogenic AA or ketones provide acetyl CoA for de novo synthesis of fatty acids (Bauman and Griinari, 2003; Zhang et al., 2015; Danes et al., 2023). Authors have shown ketones and EAA have a regulatory role on lipogenic transcription factor expression (Zhang et al., 2015; Li et al., 2016) which might have contributed to the observed differences in milk fat yield.

Study Limitations

Consideration of the application of the described feeding strategies on commercial dairy farms will require acknowledgment of the diet formulation and study limitations. Metabolizable protein supply was manipulated using 4 protein sources in the current study. The digestibility, AA composition and bioavailability as well as palatability of protein sources might affect performance responses to the described feeding strategies and should be considered when formulating diets using different ingredients. Despite the use of grouping and feeding strategies typical to North America, cows in the current study were housed in tie stalls and treatments were applied from -28 d before expected calving to 21 DIM. Grouping strategies and pen size that could affect length of diet exposure and introduce competition at the feed bunk deserve consideration. Moreover, although prepartum diets

Table 6. Effect of increasing MP supply in the prepartum, postpartum, or both diets on milk yield and composition

Variable ³	Treatment ¹				P-value ²			
	CC (n = 23)	CH (n = 24)	HC (n = 22)	HH (n = 23)	Trt	Wk	Parity	Trt x Wk
1 to 21 DIM								
Milk Yield, kg/d	39.2 ± 1.0 ^{bc}	42.4 ± 0.9 ^{ab}	38.0 ± 1.0 ^c	44.7 ± 1.0 ^a	<0.01	<0.01	<0.01	<0.01
ECM, kg/d	46.4 ± 1.1 ^{bc}	50.3 ± 1.1 ^{ab}	45.2 ± 1.1 ^c	53.6 ± 1.1 ^a	<0.01	<0.01	<0.01	<0.01
Fat, %	4.81 ± 0.11	4.82 ± 0.11	4.84 ± 0.12	4.96 ± 0.11	0.79	<0.01	0.61	0.26
Protein, %	3.20 ± 0.04	3.20 ± 0.04	3.25 ± 0.04	3.11 ± 0.04	0.15	<0.01	0.95	0.16
Lactose, %	4.76 ± 0.02 ^b	4.81 ± 0.02 ^a	4.81 ± 0.02 ^{ab}	4.79 ± 0.02 ^{ab}	0.04	<0.01	0.42	0.88
Total Solids, %	13.77 ± 0.14	13.84 ± 0.13	13.90 ± 0.14	13.87 ± 0.13	0.90	<0.01	0.61	0.81
MUN, mg/dL	6.37 ± 0.36 ^b	8.83 ± 0.35 ^a	6.92 ± 0.36 ^b	9.33 ± 0.35 ^a	<0.01	0.30	0.32	0.38
Fat yield, kg/d	1.86 ± 0.06 ^{bc}	2.03 ± 0.05 ^{ab}	1.81 ± 0.06 ^c	2.21 ± 0.06 ^a	<0.01	<0.01	<0.01	<0.01
Protein yield, kg/d	1.24 ± 0.03 ^{bc}	1.33 ± 0.03 ^{ab}	1.22 ± 0.03 ^c	1.37 ± 0.03 ^a	<0.01	<0.01	<0.01	<0.01
Lactose yield, kg/d	1.86 ± 0.05 ^b	2.04 ± 0.05 ^a	1.83 ± 0.05 ^b	2.14 ± 0.05 ^a	<0.01	<0.01	0.01	<0.01
Total Solids, kg/d	5.36 ± 0.12 ^b	5.82 ± 0.12 ^a	5.24 ± 0.12 ^b	6.17 ± 0.12 ^a	<0.01	<0.01	<0.01	<0.01
Linear Score	2.76 ± 0.32	3.20 ± 0.31	2.28 ± 0.33	2.27 ± 0.32	0.12	<0.01	0.09	0.82
ECM/DMI	2.36 ± 0.06 ^{ab}	2.32 ± 0.05 ^b	2.28 ± 0.06 ^b	2.57 ± 0.06 ^a	<0.01	<0.01	0.64	0.18
Milk N:Feed N	0.42 ± 0.01 ^a	0.38 ± 0.01 ^b	0.41 ± 0.01 ^a	0.41 ± 0.01 ^{ab}	<0.01	<0.01	0.28	0.66
22 to 42 DIM								
Milk Yield, kg/d	49.6 ± 1.0 ^b	53.3 ± 1.0 ^a	49.3 ± 1.0 ^b	54.1 ± 1.0 ^a	<0.01	<0.01	<0.01	0.21
ECM, kg/d	53.1 ± 1.1 ^b	58.3 ± 1.1 ^a	53.1 ± 1.1 ^b	58.0 ± 1.1 ^a	<0.01	<0.01	<0.01	0.53
Fat, %	4.09 ± 0.09	4.28 ± 0.09	4.10 ± 0.10	4.11 ± 0.09	0.38	<0.01	0.08	0.62
Protein, %	2.83 ± 0.03	2.82 ± 0.03	2.91 ± 0.03	2.79 ± 0.03	0.10	0.02	0.13	0.08
Lactose, %	4.85 ± 0.01	4.86 ± 0.01	4.88 ± 0.01	4.85 ± 0.01	0.42	<0.01	0.76	0.69
Total Solids, %	12.69 ± 0.11	12.89 ± 0.11	12.84 ± 0.11	12.70 ± 0.11	0.45	<0.01	0.22	0.41
MUN, mg/dL	8.96 ± 0.31	9.61 ± 0.31	9.46 ± 0.32	9.86 ± 0.32	0.15	<0.01	0.79	0.07
Fat yield, kg/d	2.02 ± 0.05 ^b	2.27 ± 0.05 ^a	2.01 ± 0.05 ^b	2.22 ± 0.05 ^a	<0.01	0.05	<0.01	0.64
Protein yield, kg/d	1.40 ± 0.03	1.50 ± 0.03	1.43 ± 0.03	1.51 ± 0.03	0.03	<0.01	<0.01	0.35
Lactose yield, kg/d	2.41 ± 0.05 ^b	2.59 ± 0.05 ^a	2.41 ± 0.05 ^b	2.62 ± 0.05 ^a	<0.01	<0.01	<0.01	0.10
Total Solids, kg/d	6.29 ± 0.12 ^b	6.86 ± 0.12 ^a	6.32 ± 0.13 ^b	6.87 ± 0.12 ^a	<0.01	<0.01	<0.01	0.49
Linear Score	1.74 ± 0.36	2.27 ± 0.35	1.34 ± 0.37	1.63 ± 0.36	0.28	0.02	0.25	0.68
ECM/DMI	2.14 ± 0.05	2.19 ± 0.05	2.13 ± 0.05	2.29 ± 0.05	0.08	<0.01	0.15	0.37
Milk N:Feed N	0.38 ± 0.01 ^a	0.35 ± 0.01 ^b	0.39 ± 0.01 ^a	0.37 ± 0.01 ^{ab}	<0.01	0.01	0.75	0.36

^{a-c} LSM with different superscripts differ ($P \leq 0.05$; Tukey's test).

¹Prepartum diets, fed starting at 28 d before expected calving, were formulated to contain Met and Lys at 1.24 and 3.84 g/Mcal metabolizable energy (ME), respectively and supply a control (C; 85 g MP/kg DM) or high (H; 113 g MP/kg DM) level of estimated MP. Postpartum diets, fed from 1 to 21 DIM, were formulated to contain Met and Lys at 1.15 and 3.16 g/Mcal ME, respectively and supply control (C; 104 g MP/kg DM) or high (H; 131 g MP/kg DM) level of estimated MP. The combination of a pre- and postpartum diet resulted in treatments: 1) CC, 2) CH, 3) HC, 4) HH. From 22 to 42 DIM, cows were fed a lactation diet that contained Met and Lys at 1.01 and 2.88 g/Mcal ME, respectively and supplied 113 g MP/kg.

²Fixed effects of treatment (trt), time (wk), parity (2 vs. ≥ 3), and the interaction of treatment x week.

³Reported as LSM $\pm$ SEM.

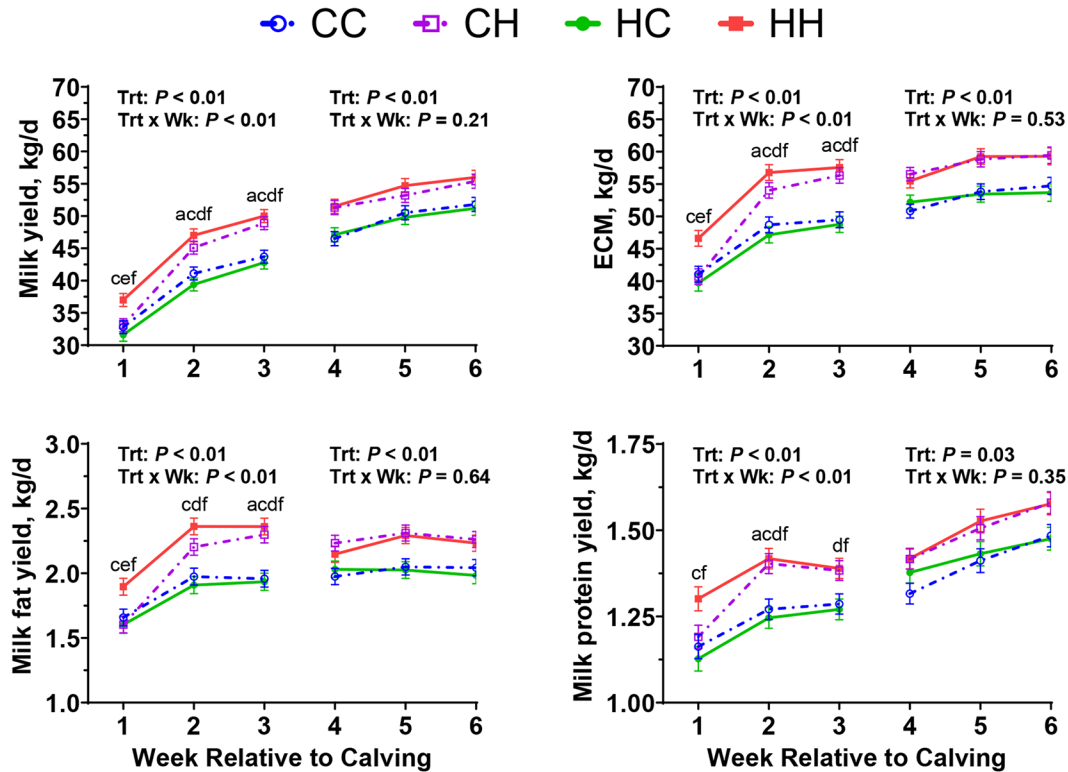


Figure 4. Effect of increasing MP supply in the prepartum, postpartum, or both diets on milk yield, ECM, milk fat yield, and milk protein yield. Prepartum diets were formulated to contain Met and Lys at 1.24 and 3.84 g/Mcal ME, respectively and supply a control (C; 85 g MP/kg DM) or high (H; 113 g MP/kg DM) level of estimated MP. Postpartum diets were formulated to contain Met and Lys at 1.15 and 3.16 g/Mcal ME, respectively and supply control (C; 104 g MP/kg DM) or high (H; 131 g MP/kg DM) level of estimated MP. The combination of a pre- and postpartum diet resulted in treatments: 1) CC, 2) CH, 3) HC, 4) HH. From 22 to 42 DIM, cows were fed a lactation diet that contained Met and Lys at 1.01 and 2.88 g/Mcal ME, respectively and supplied 113 g MP/kg. If the interaction term resulted in $P \leq 0.05$, letters denote differences ($P \leq 0.05$; Tukey's test) between treatments at individual time point. a = CC vs. CH, b = CC vs. HC, c = CC vs. HH, d = CH vs. HC, e = CH vs. HH, f = HC vs. HH. Data presented as LSM $\pm$ SEM P -values for all main effects are in Table 6.

were formulated to contain a similar DCAD, the chemical analysis of diets resulted in a difference in DCAD between prepartum diets and should be acknowledged as a study limitation. In addition, primiparous cows have been shown to benefit from additional protein (Husnain and Santos, 2019; NASEM, 2021) and the effect of these dietary strategies on primiparous cows would require further investigation as we did not include nulliparous heifers in our study.

CONCLUSIONS

Increasing prepartum MP supply, with diets formulated such that Met and Lys were matched on an ME basis, increased prepartum BW gain but did not affect relative DMI or BCS. During wk 1, cows fed a high level of MP during both the pre- and postpartum period had the highest milk yield. From wk 2 on, milk and components yields were greatest in cows fed a greater postpartum MP supply, regardless of prepartum treatment. Metabolizable protein supply did not affect DMI, hyperketonemia

events, NEFA concentrations, or BCS loss from 1 to 21 DIM. From 22 DIM to the end of the observational period at 42 DIM, cows fed a greater postpartum MP supply had increased milk and ECM yields suggesting diet strategy can affect lactation performance beyond the initial feeding period. Findings from this study suggest that multiparous cows respond to a postpartum diet with increased MP supply and formulated with adequate levels of Met and Lys relative to ME with increased milk yield.

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Table 7. Effect of increasing MP supply in the prepartum, postpartum, or both diets on postpartum plasma metabolic indicators

Variable ³	Treatment ¹				P-value ²			
	CC (n = 23)	CH (n = 24)	HC (n = 22)	HH (n = 23)	Trt	Time	Parity	Trt x time
1 to 21 DIM								
Glucose, mg/dL	55.1 ± 1.1 ^{ab}	56.9 ± 1.1 ^{ab}	57.2 ± 1.2 ^a	53.1 ± 1.1 ^b	0.03	<0.01	0.75	0.70
NEFA, µEq/L	591 (516–676)	521 (457–595)	572 (498–657)	600 (524–687)	0.41	<0.01	0.40	0.36
BHB, mmol/L	0.76 (0.65–0.88) ^{ab}	0.83 (0.72–0.96) ^a	0.60 (0.51–0.69) ^b	0.90 (0.78–1.05) ^a	<0.01	<0.01	0.22	0.50
PUN, mg/dL	4.3 (4.0–4.6) ^c	6.8 (6.3–7.3) ^a	4.8 (4.5–5.2) ^b	7.5 (7.0–8.1) ^a	<0.01	0.02	0.21	0.02
Hyperketonemia event ⁴	9/69 (13.0%)	7/72 (9.7%)	2/66 (3.0%)	10/69 (14.5%)	0.13			
22 to 42 DIM								
Glucose, mg/dL	59.5 ± 1.1	59.2 ± 1.1	61.7 ± 1.1	59.0 ± 1.1	0.30	<0.01	0.75	0.05
NEFA, µEq/L	445 (371–534) ^{ab}	375 (314–448) ^{ab}	356 (296–428) ^b	495 (413–594) ^a	0.02	<0.01	0.85	0.52
BHB, mmol/L	0.74 (0.62–0.88) ^{ab}	0.82 (0.69–0.97) ^a	0.58 (0.49–0.70) ^b	0.87 (0.73–1.04) ^a	<0.01	<0.01	0.21	0.60
PUN, mg/dL	6.9 (6.3–7.4)	7.5 (6.9–8.1)	6.9 (6.3–7.4)	7.8 (7.2–8.5)	0.04	<0.01	0.97	<0.01

^{a-c} LSM with different superscripts differ ($P \leq 0.05$; Tukey's test).

¹Prepartum diets, fed starting at 28 d before expected calving, were formulated to contain Met and Lys at 1.24 and 3.84 g/Mcal metabolizable energy (ME), respectively and supply a control (C; 85 g MP/kg DM) or high (H; 113 g MP/kg DM) level of estimated MP. Postpartum diets, fed from 1 to 21 DIM, were formulated to contain Met and Lys at 1.15 and 3.16 g/Mcal ME, respectively and supply control (C; 104 g MP/kg DM) or high (H; 131 g MP/kg DM) level of estimated MP. The combination of a pre- and postpartum diet resulted in treatments: 1) CC, 2) CH, 3) HC, 4) HH. From 22 to 42 DIM, cows were fed a lactation diet that contained Met and Lys at 1.01 and 2.88 g/Mcal ME, respectively and supplied 113 g MP/kg.

²Fixed effects of treatment (trt), time, parity (2 vs. ≥ 3), and the interaction of treatment x time.

³Plasma concentrations of glucose, nonesterified fatty acids (NEFA), BHB, and plasma urea nitrogen (PUN) were determined twice weekly from 1 to 28 DIM and once weekly from 29 to 42 DIM. Reported as LSM $\pm$ SEM or back-transformed LSM (95% confidence interval).

⁴BHB ≥ 1.2 mmol/L from 3 to 10 DIM.

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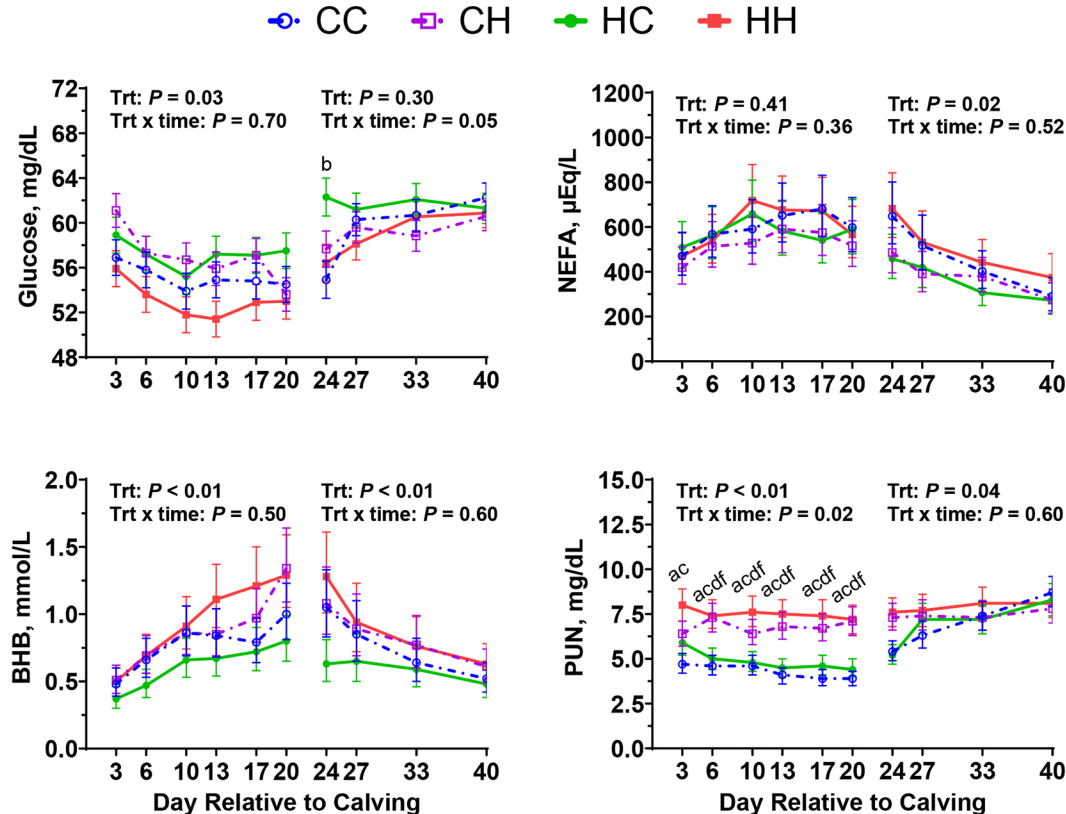


Figure 5. Effect of increasing MP supply in the prepartum, postpartum, or both diets on concentrations of glucose, nonesterified fatty acids (NEFA), BHB, and plasma urea nitrogen (PUN). Prepartum diets were formulated to contain Met and Lys at 1.24 and 3.84 g/Mcal ME, respectively and supply a control (C; 85 g MP/kg DM) or high (H; 113 g MP/kg DM) level of estimated MP. Postpartum diets were formulated to contain Met and Lys at 1.15 and 3.16 g/Mcal ME, respectively and supply control (C; 104 g MP/kg DM) or high (H; 131 g MP/kg DM) level of estimated MP. The combination of a pre- and postpartum diet resulted in treatments: 1) CC, 2) CH, 3) HC, 4) HH. From 22 to 42 DIM, cows were fed a lactation diet that contained Met and Lys at 1.01 and 2.88 g/Mcal ME, respectively and supplied 113 g MP/kg. If the interaction term resulted in $P \leq 0.05$, letters denote differences ($P \leq 0.05$; Tukey's test) between treatments at individual time point. a = CC vs. CH, b = CC vs. HH, c = CC vs. HH, d = CH vs. HC, e = CH vs. HH, f = HC vs. HH. Data presented as LSM $\pm$ SEM or back-transformed LSM and 95% confidence interval. P -values for all main effects are in Table 7.

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